Herded or Anchored? Testing Conformity in Machine Learning on a Decade of ICLR Submissions

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Abstract

Machine learning is often said to herd: researchers chase what is hot, and review favours it. But a field can converge without copying. Researchers may instead stay close to a record: their own earlier work, or a topic they already know. We separate these two routes to conformity, imitation and continuity, and test the herding account where they predict different things. The data are 43,480 ICLR submissions (2017–2026), rejected and withdrawn papers included, with hypotheses registered before outcomes were read and re-tested on 11,099 held-out papers. The field did converge on foundation models, but convergence fits both routes, and how many topics look emergent depends on hindsight: 8 to 29, depending on the years a topic map is fitted on. Where the routes part, no herding is detectable. Researchers new to a topic propose methods more often than incumbents do, the reverse of our registered prediction. At review, crowded topics show neither a detectable premium nor a penalty, while papers far from their authors' earlier work are accepted less often (post hoc, confirmed on held-out papers). The conformity we can detect at ICLR points to a record, not to the crowd.

1 Introduction

Machine learning is often said to herd. Between 2017 and 2026, yearly submissions to the International Conference on Learning Representations (ICLR) in our discovery corpus grew 40-fold, from 384 to 15,411. The share building on a foundation model (FM) rose from 0.3% to 54.4%, and the share proposing a new architecture fell from 24.5% to 7.5%. Keywords rose and fell (Figure 1): generative adversarial networks fell from 8.1% of submissions in 2018, and large language models reached 16.7% in 2026. The wave is NLP's too, as its researchers increasingly publish at ML venues (Jurgens, 2026). Whether science has fashions is an old question (Crane, 1969). Critics of ML scholarship point to incentives that favour short-term success (Lipton and Steinhardt, 2019) and to convergence on what suits the field's hardware, benchmarks and foundation models (Hooker, 2021; Dehghani et al., 2021; Bommasani et al., 2021); cascade and fashion theories explain waves by people taking their cue from others (Bikhchandani et al., 1992; Banerjee, 1992; Abrahamson, 1991).

Is the switch to FMs not herding enough? Convergence on a shared substrate is predicted by lock-in, by the efficient choice of a tool that works and by a bandwagon alike, so the curve cannot tell them apart. The herding question is about behaviour: what researchers write when they enter a topic, and what the review gate accepts. There, two routes to conformity part. *Imitation* takes its reference point from what others are doing now: what is popular, rising or recently accepted. *Continuity* takes it from a record: the team's own earlier work, or a topic that a team member already knows. Imitation makes teams alike; continuity keeps each team near its own past. We call work that conforms by continuity *anchored*.[1]

Read as herding, the waves make three claims: they are fashions; newcomers to a topic carry them by following others; and review favours following. We test each claim at its site, the field, entry to a topic and the acceptance gate, on 43,480 ICLR submissions from 2017–2026, rejected and withdrawn papers included. Hypotheses were registered before any full-corpus label existed, outcomes stayed sealed until one frozen run, and the main tests were run again on 11,099 held-out papers: a pre-registered out-of-sample replication, not a certified untouched holdout. Each finding carries one of three labels: REPLICATED (from a registered test,

[1] 'Anchored' names a position relative to a record, not a mechanism. It is not the anchoring-and-adjustment heuristic (Tversky and Kahneman, 1974), the sense in which writing on herding in science uses the word (Baddeley, 2015).

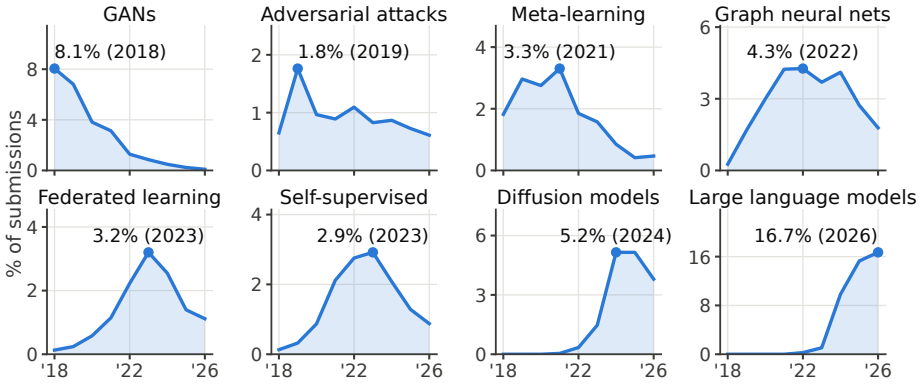


Figure 1: Research waves at ICLR: share of discovery submissions listing each author keyword, 2018–2026; dots mark the peak. The eight keywords were chosen for display after their curves were known. Across the 68 most frequent keywords, 17 peaked in 2019–2024 with shares of at most half the peak both before it and in 2026. Uses no acceptance data.

supported again on held-out papers), HELD-OUT SUPPORTED (found post hoc, then supported by a test declared before the held-out data were read) or EXPLORATORY (discovery only, or descriptive). Tests without a verdict are called inconclusive or uninformative (Section 3).

At the field level, keyword waves are real, but emergence depends on hindsight, and the durable shift, the FM package, fits every route. At entry, researchers new to a topic propose methods more often than incumbents, while, in the FM era, benchmark and evaluation work sits with incumbents: the opposite of the follow-on entry that herding implies, and of our own registered prediction. At the gate, no premium for crowded topics is detectable, the branded title's small premium lies below what the held-out test could detect, and what replicated, apart from regressing topic edges, is tied to the team's own record.

Prior work relates ICLR acceptance to prestige, process and research area (Tran et al., 2020; Ibrahim et al., 2026; Jung et al., 2025; Wang et al., 2023; Xu et al., 2026) and describes which papers start topics, who enters LLM research and how far researchers move from their own past work (Huang, 2026; Movva et al., 2024; Bao and Li, 2025). To our knowledge, among the works we checked, none tests the herding account at entry and at the gate within one submission population, with registered hypotheses re-tested on held-out papers, as Ibrahim et al. (2026) did for a prestige gap. We contribute:

- **Entry.** A registered test of herding where imitation predicts it most sharply. Researchers entering a topic propose methods more often (+3.7 pp; REPLICATED). The surplus holds for researchers with ICLR work in other topics and predates the FM wave, while, in the FM era, benchmark and evaluation work sits with topic incumbents and larger teams (EXPLORATORY).
- **Gate.** Using every decided submission under double-blind review, rejected ones included, acceptance goes with closeness to the team's own record among teams that have one (far jump -5.4 pp, held-out -6.1; HELD-OUT SUPPORTED), not detectably with the crowd. Hill et al. (2025) relate pivots to publication success; we add the accept/reject decision itself.
- **Method.** On the same papers, the number of emerging topics ranges from 8 to 29 depending on the years the topic map is fitted on (EXPLORATORY). Emergence is hard to judge before the fact (Rotolo et al., 2015; Li et al., 2025), and a prospective signal has been tested (Khanbayov and Kurban, 2026); an early-window refit is a cheap check on trend claims.

2 Two Routes to Conformity

Imitation. When outcomes are uncertain, organisations mimic one another (DiMaggio and Powell, 1983), individuals follow their predecessors and discard their own information (Bikhchandani et al., 1992; Banerjee, 1992), and herding protects a reputation when evaluators judge people against their peers (Scharfstein and Stein, 1990); bandwagons can spread even when adopting brings no gain (Abrahamson and Rosenkopf, 1993). In ML, benchmark use concentrates on a small set of datasets, increasingly so (Koch et al., 2021). Entrants to a topic face the most uncertainty, so imitation predicts that they flock to hot topics, adopt a rising package

Route (reference point)	Field	Entry	Gate
Imitation (others, now)	Convergence on a package	Entrants flock to hot topics, adopt the rising package early and write follow-on benchmark and evaluation work	A premium for crowded topics, the popular form and riding a wave
Continuity (a record)	Convergence on a package	Entrants bring other work; incumbents consolidate; the package travels with a record	Acceptance tracks closeness to the team's own record

Table 1: What each route predicts at the three sites. Lock-in and efficient choice also predict convergence on a package, so only entry and the gate can tell the routes apart. Imitation's entry and gate predictions were registered before outcomes were read; continuity's were written after the tests had run.

early, and write follow-on work that benchmarks or evaluates what others proposed. At the gate, if review favours following, papers in crowded topics, in the popular form or riding a wave are accepted more often.

Continuity. Search stays near existing competence (March, 1991; Stuart and Podolny, 1996), and taking up new knowledge depends on related prior knowledge (Cohen and Levinthal, 1990). At entry, continuity predicts that incumbents consolidate a topic's record and that a new package travels with a record; on outsider accounts, entrants bring work from elsewhere (Zeng et al., 2021; Packalen and Bhattacharya, 2019). At the gate, acceptance should track how close a paper stays to its team's record. Weisberg and Muldoon (2009) separate 'followers' of others' approaches from agents who ignore them; we borrow the distinction, not their welfare conclusions (Alexander et al., 2015).

Rivals, and when the predictions were written. Visible status, cumulative advantage and expertise (Merton, 1968; Ibrahim et al., 2026; Li, 2017) also predict that acceptance goes with a team's record. Network diffusion, in which less connected actors adopt later (Coleman et al., 1957), also predicts that teams new to ICLR take up a package later, so that contrast cannot separate the routes. Imitation's entry and gate predictions (Table 1) are the herding hypotheses we registered before reading any outcome; continuity's were written after the tests had run. The distinction organises the tests; it did not generate them. To our knowledge, among the works we checked, the two routes have not been tested against each other at entry or at the gate.

3 Data and Design

Corpus. The corpus comes from our own dated acquisitions of official OpenReview records (Appendix A): 43,480 discovery submissions with a usable abstract and 11,099 held-out papers. Acceptance tests use the 32,190 decided discovery papers from 2018–2026, 40.1% of them accepted, and 8,185 decided held-out papers from 2018–2026; withdrawn papers are left out.

Instruments. A *topic* is one of 100 spherical k-means clusters of SPECTER2 embeddings of title and abstract (Cohan et al., 2020; Singh et al., 2023), fitted on all years unless another map is named. An LLM with a fixed codebook gives each paper one primary contribution type: a *method paper* proposes a method, including a new architecture, and a *measurement paper* contributes a dataset, benchmark or evaluation; an *FM paper* uses or adapts a large pre-trained model. A blind second pass agreed at κ 0.93, and the author hand-checked a small random sample. This is consistency between LLM labellings, not human-level validity; zero-shot LLM labels reach only fair agreement with human coders on many tasks (Ziems et al., 2024).

Entry profiles. Entry status is relative to a topic and counts ICLR submissions only. In a *newcomer paper*, no author has an earlier ICLR submission, with any outcome, in the paper's topic (a topic-relative definition, as in Biliotti et al., 2024); otherwise it is an *incumbent paper*, and an author with such a submission is a *topic insider*. A newcomer paper comes from a *first-time team* if no author has any earlier ICLR submission, and from a *returning topic newcomer* if some author has one in another topic.

Two distances. Both distances use one centred SPECTER2 space, with different reference points. *Closeness to the team's own record*, $F_{\max}$, is the highest cosine between the paper and the mean of an author's earlier ICLR submissions since 2021, taken over the paper's *experienced authors*, those with such submissions (following Bao and Li, 2025). *Distance from earlier ICLR work* is one minus the mean cosine to the ten nearest submissions of earlier years, as a percentile within topic and year.

Prospective measurement. Predictors use only the past: *prior FM adoption* is the FM share of the topic's previous-year papers, *crowding* is the

log of one plus the topic's earlier papers, and a *hot topic* is in the top fifth of prior growth. Code links leak the outcome into 17.6% of accepted but 0.5% of rejected 2023–2026 abstracts (Kaufman et al., 2012); in discovery, the distance-based gate estimates hold without them (no link-removed text exists for held-out papers; Appendix F).

Protocol. Hypotheses, models and a support rule were registered before any full-corpus label existed. A hypothesis is supported if it has the predicted sign, a Holm-adjusted $p < 0.05$ in its family (Holm, 1979), and that sign in at least 75% of counted variants and in both hindsight variants (a year-balanced map; the 2017–2020 map applied to 2021–2026 papers). Outcomes could be read only by committed code in one logged, gated run (Nosek et al., 2018; van den Akker et al., 2021). Post hoc analyses were declared before they were run, and their one-sided held-out tests before any held-out estimate for that question was computed. Models are linear probability models with topic and year fixed effects and topic-clustered errors. We disclose two weaknesses. Under a simulated null whose structure cuts across topics, the main entry tests rejected 12.6% and 10.0% of the time at a nominal 5%, which a wild cluster bootstrap (Cameron et al., 2008) did not fix (Appendix H). And keyword popularity and title style had been explored against acceptance on the discovery papers before registration: the discovery estimate for titles is not a test, and the keyword-timing tests probe a dimension already seen. A test that is not supported and whose estimate lies below its minimum detectable effect at 80% power (MDE80) is *uninformative*: it shows neither an association nor its absence. Supported held-out tests confirm a direction, not a size.

4 The Field: What Is Fashion and What Lasts

Keyword waves are real, but how many topics look emergent depends on hindsight, and the one durable shift, the FM package, fits every route.

Waves. Author keywords need no topic map, and at their level the waves are real: of the 68 most frequent keywords, 17 rose and fell, peaking in 2019–2024 with shares of at most half the peak both before it and in 2026 (Figure 1; EXPLORATORY).

Emergence depends on hindsight. Histories of the field usually fit one topic model to the whole

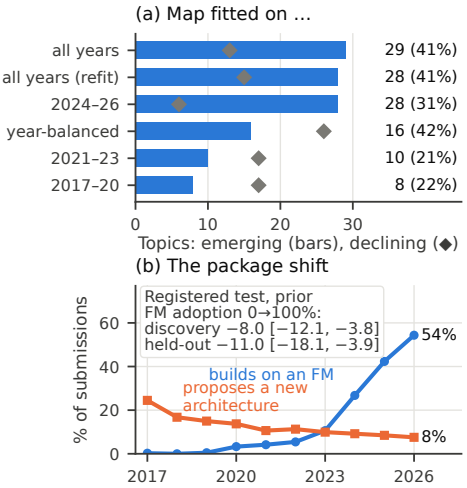


Figure 2: (a) Emerging (bars) and declining (grey diamonds) topics on $k = 100$ maps fitted on different years, same 43,480 papers; right-hand labels: emerging count (share of 2024–2026 papers in emerging topics, 22.3% on the 2017–2020 map against 40.9% on the all-years map). All years (refit): the all-years map refitted with another seed. Topic size 'predicts' two-year growth only on the all-years map (R^2 0.28 against −0.03; no paired test). (b) Share of submissions building on an FM (blue circles) and proposing a new architecture (orange squares); LLM labels. Inset: the registered test of the package shift, REPLICATED (held-out confirms direction, not size). Uses no acceptance data.

archive (Hall et al., 2008; Anderson et al., 2012; Klinger et al., 2020), and dynamic topic models and BERTopic fit all years at once (Blei and Lafferty, 2006; Grootendorst, 2022). We call a topic *emerging* if its 2024–2026 share is at least three times its 2017–2020 share. On the same papers and pipeline, a map fitted on 2017–2020 finds 8 emerging topics and a map fitted on all years finds 29 (Figure 2a, Appendix C; EXPLORATORY). Two reasons combine, and we cannot separate them. A map fitted on all years leaks the future into the past, like temporal leakage in prediction (Kapoor and Narayanan, 2023) and hindsight in judgement (Fischhoff, 1975): with 69% of papers from 2024–2026, it is in effect a present-day map. And the early map cannot represent topics born after 2020. The two counts therefore bracket the number of emerging topics rather than fix it.

What lasts: a package. The FM share rose every year from 2018 (Figure 2b). Where more of a topic's previous-year papers built on FMs, fewer of its papers proposed new architectures: −7.98 pp [−12.14, −3.82] for prior adoption from 0 to 100% in discovery and −11.0 [−18.1, −3.9] on held-out

papers (REPLICATED; the held-out test confirms the direction, not the size). No detectable shift towards empirical or evaluation papers came with it (−0.63 [−3.64, +2.38]; not supported, inconclusive), and the association may also run the other way. The pattern matches accounts of NLP's paradigm shifts (Gururaja et al., 2023; Saphra et al., 2024; Pramanick et al., 2023). But lock-in (Hooker, 2021), efficient choice (Sutton, 2019) and a bandwagon (Fujimura, 1988) all predict such convergence, so the field level cannot tell imitation from continuity. Entry and the gate must.

5 Entry: Who Writes What

Entry shows a division of labour: researchers entering a topic bring methods, while, in the FM era, benchmark and evaluation work sits with larger teams and topic incumbents. This is the opposite of the follow-on entry that the herding account implies.

Newcomers propose methods. We registered that newcomer papers, whose authors face the most uncertainty, would more often be follow-on work: benchmarks, datasets and evaluations of what others had proposed. Newcomer papers are 72.9% of the 42,325 papers from 2019–2026 with an author identifier. They are more often method papers, +3.71 pp [+2.61, +4.80] in discovery and +3.60 [+1.48, +5.71] held-out, and less often measurement papers, −1.70 [−2.46, −0.93] and −2.10 [−3.35, −0.85] (Table 2, Figure 3; REPLICATED). Our registered prediction thus failed in the opposite direction, and the reversed hypotheses, registered before any held-out data existed, held on held-out papers. All 10 counted variants agree, and the t-statistics (6.7 and −4.4) are large against the inflated test size (Appendix H). Much of the measurement gap is accounted for by team size: with log team size it shrinks to −0.54 [−1.11, +0.04] in discovery and −0.98 held-out, while the method gap does not move (+3.92 and +3.64).

Entering a topic, not ICLR. Newcomer papers split into first-time teams (26.3% of papers; median team of 3) and returning topic newcomers (46.7%; median 5); incumbent papers are the other 27.0% (median 5). Returning topic newcomers write method papers more often than incumbents, +3.43 pp [+2.31, +4.55], at the same median team size and also with team size controlled (+3.62). First-time teams do so by +4.35 [+2.82, +5.89], and the two newcomer profiles do not differ detectably (+0.92 [−0.44, +2.29]; all EXPLORATORY). The method surplus thus goes mainly with entering a topic, not with being new to ICLR or with small teams, and it is present in every era, so it predates the FM wave (Appendix E). The measurement gap is different. For first-time teams it is accounted for by team size (−2.24 [−3.24, −1.25] against incumbents, +0.09 [−0.63, +0.80] with team size); for returning topic newcomers part of it remains (−1.47, and −0.81 [−1.46, −0.16] with team size). It appears only in 2024–2026 (first-time teams against incumbents −2.75 [−4.07, −1.44]). In the FM era, then, benchmark and evaluation work sits with topic incumbents and larger teams; team size is the only resource we measure. NLP shows a parallel at the level of research trends, where tasks and methods lead and datasets follow (Pramanick et al., 2023).

Contrast	Discovery [95% CI]	Held-out [95% CI]	Label
Method paper	+3.71 [+2.61, +4.80]	+3.60 [+1.48, +5.71]	REPLICATED
Measurement paper	−1.70 [−2.46, −0.93]	−2.10 [−3.35, −0.85]	REPLICATED
Measurement paper, with log team size[a]	−0.54 [−1.11, +0.04]	−0.98	registered variant
Method: returning topic newcomer − incumbent[b]	+3.43 [+2.31, +4.55]	–	EXPLORATORY
Method: first-time team − returning topic newcomer[b]	+0.92 [−0.44, +2.29]	–	EXPLORATORY
Distance from earlier ICLR work, own work excluded (percentile points)	+4.33 [+3.44, +5.22]	–	EXPLORATORY
Flocking: incumbent share, hot − other topics	−0.73 [−2.68, +1.22]	not registered	inconclusive

Table 2: Entry: newcomer minus incumbent papers unless the row names other profiles; pp within topic and year. Held-out tests are one-sided and confirm direction, not size. [a]Registered supplementary variant, outside the support rule. [b]First-time team: no author has any earlier ICLR submission; returning topic newcomer: an author has one, in another topic (Appendix E).

Farther from earlier work. Newcomer papers also sit farther from earlier ICLR work: +5.90 percentile points [+4.95, +6.85] within topic and year, +5.24 among method papers alone, and +4.33 [+3.44, +5.22] with the authors' own earlier papers excluded from the reference set (EXPLORATORY). Papers by topic newcomers with an ICLR record are also less likely than incumbent papers to have a same-year *twin*, an independent submission at cosine ≥ 0.70 (with first-time teams con-

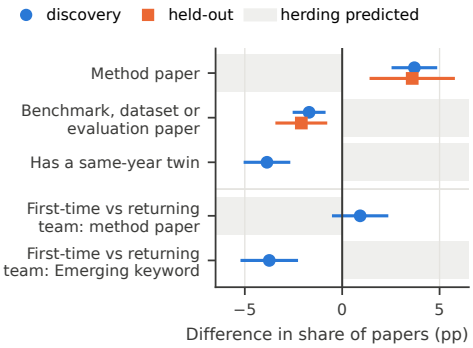


Figure 3: Entry contrasts within topic and year (pp) with 95% intervals: discovery (blue circles) and held-out (orange squares). Rows 1–2, newcomer minus incumbent papers, are registered and REPLICATED (held-out confirms direction, not size). Rows 3–5 are EXPLORATORY (discovery only): row 3 compares papers by topic newcomers with an ICLR record with incumbent papers (first-time teams controlled separately; class, FM and team-size controls); row 4 compares first-time teams with returning topic newcomers, and row 5 with all teams that have an ICLR record. Grey: the side the herding prediction expected. Uses no acceptance data.

trolled separately): −3.86 pp [−4.98, −2.74] (EXPLORATORY). The follow-on prediction expects newcomer papers to sit closer to earlier work, not farther. Writing method papers could itself imitate the venue's dominant genre, three quarters of ICLR papers, but newcomers' distance from earlier work tells against that reading. A method paper is a contribution type, and embedding distance is not originality: these measures give no sign that newcomers follow, and do not show that they are more original.

Later onto the package. In 2023–2026 topics at 5–50% prior FM adoption, first-time teams write FM papers less often than returning topic newcomers, by −4.58 pp [−6.55, −2.60] up to −14.71 [−19.82, −9.61] as adoption rises, while returning topic newcomers do not differ detectably from incumbents (Figure 5 in Appendix E). First-time teams also list young, rising *Emerging* keywords less often than teams with an ICLR record: −3.74 [−5.14, −2.35] (both EXPLORATORY). The package travels with an ICLR record, as absorptive capacity predicts (Cohen and Levinthal, 1990); network diffusion predicts the same order, so this contrast does not discriminate.

What stays open. Whether entrants crowd into hot topics is unsettled: the registered flocking test, the incumbent share in hot against other topics, is inconclusive (−0.73 [−2.68, +1.22]). At entry, no detectable difference fits the follow-on prediction.

6 The Gate: Crowd or Record?

At the gate we detect no premium for following the crowd. Apart from topic edges, which regress, what replicated is tied to the team's own record, which continuity predicts and status and expertise predict too.

Does acceptance go with following the crowd? The registered crowding test predicted a penalty, and it is uninformative: −1.68 pp per log unit (natural log of one plus the topic's earlier papers) [−3.68, +0.32] in discovery and −1.51 [−5.20, +2.19] held-out, against an MDE80 of 4.8 (Table 3, Figure 4). Neither a penalty nor a premium is detectable, and held-out premiums above +2.19 pp per log unit lie outside the interval.

The plainest imitation-like spread in the corpus is in form. The share of titles that open with a short coined name and a colon rose from 10.9% in 2017 to 44.6% in 2026 (Appendix G). Within topic and year, with class, FM and team-size controls, such branded titles are accepted +1.74 pp [+0.37, +3.11] more often in discovery, which is not a test, since title style had been explored before registration. The declared held-out test gives +1.76 [−0.82, +4.34] (one-sided p 0.089, MDE80 3.2): not supported, and uninformative, so the premium stays EXPLORATORY. Raw within-year gaps are larger (+3.3 in discovery, +4.0 held-out); about half goes with topic, type, FM use and team size. Titles can also be revised after the decision, which could create such a premium.

FM papers are accepted more often, +4.62 pp [+2.79, +6.44], with no detectable difference between late and early adopters: the premium changes by +0.19 [−0.96, +1.34] per logit unit (log-odds of the topic's prior FM adoption; EXPLORATORY), which fits rewarded imitation and efficient choice alike. A topic's *edge*, its acceptance minus that of all other papers of the same year (topics of the 2017–2020 map), is real but regresses: the 2018–2022 edge predicts the 2024–2026 edge with slope 0.38 [0.18, 0.58] among discovery papers, and the held-out papers' 2024–2026 edges regress on the discovery 2018–2022 edges with slope 0.60 [0.32, 0.88] (HELD-OUT SUPPORTED; two different comparisons). A favoured topic keeps part of its edge, not all of it, as expected when decisions carry a large arbitrary component (Cortes and Lawrence,

Acceptance and …	Discovery [95% CI]	Held-out [95% CI]	Label
Others: the crowd, the form, the package			
Crowding, per log unit[a]	−1.68 [−3.68, +0.32]	−1.51 [−5.20, +2.19]	uninformative
Branded title[b]	+1.74 [+0.37, +3.11]	+1.76 [−0.82, +4.34]	EXPLORATORY
FM premium, per logit unit of prior adoption	+0.19 [−0.96, +1.34]	–	EXPLORATORY
Topic edge 2018–22 → 2024–26, slope[c]	0.38 [0.18, 0.58]	0.60 [0.32, 0.88]	HELD-OUT SUPPORTED
Emerging − Mainstream keyword[b,c]	+2.67 [+0.76, +4.58]	+3.25 [−0.18, +6.69]	EXPLORATORY
The team's own record			
Far jump: $F_{\max}$ p10 − median	−5.44 [−7.10, −3.78]	−6.09 [−9.32, −2.86]	HELD-OUT SUPPORTED
Same author, +0.1 cosine closer[c]	+0.55 [+0.21, +0.90]	+1.06 [+0.23, +1.89]	HELD-OUT SUPPORTED
One topic insider vs none[c]	+3.42 [+1.18, +5.66]	+3.88 [+0.28, +7.47]	HELD-OUT SUPPORTED
The field's record			
Least-distant fifth − middle fifth	−3.27 [−4.93, −1.60]	−0.13 [−3.50, +3.25]	uninformative

Table 3: The gate: pp within topic and year (topic edges and keywords: against the same year's other papers). Rows differ in population and unit: compare replication status, not size. Held-out tests are one-sided and confirm direction, not size. [a]Registered; predicted a penalty; MDE80 4.8. [b]Explored against acceptance before registration. Titles: the discovery estimate is not a test; held-out MDE80 3.2. Keywords: the discovery estimate is a declared post hoc test on a dimension already seen. [c]Holm across the six secondary held-out tests: topic edge 0.019, same author 0.026, insider 0.052, Emerging keywords 0.063.

2021; Beygelzimer et al., 2023). Keyword timing is weak (EXPLORATORY; Appendix D): on held-out papers, *Emerging* keywords go with higher acceptance than old, flat *Mainstream* ones (+3.25 [−0.18, +6.69]), and acceptance may fall after a keyword's share peaks (−5.2 [−11.1, +0.7]; −1.7 without the six keywords inspected first). Neither shows the flat premium that rewarded imitation of what is popular predicts.

What does acceptance go with? The clearest pattern is the *far jump*. Among 20,430 decided 2023–2026 papers with an experienced author, acceptance at the 10th percentile of closeness to the team's own record minus that at the median is −5.44 pp [−7.10, −3.78] in discovery and −6.09 [−9.32, −2.86] held-out, with power 0.89 (HELD-OUT SUPPORTED; the declared primary test). Adjusted acceptance rises with closeness up to the fourth fifth (Figure 6 in Appendix F). The far jump holds with a newcomer control (−5.0), although closeness correlates −0.57 with the newcomer flag. Holding the author fixed, a paper 0.1 closer in cosine to that author's earlier work goes with acceptance +0.55 pp [+0.21, +0.90] higher in discovery and +1.06 [+0.23, +1.89] held-out (HELD-OUT SUPPORTED; it survives a family-wise correction across the secondary tests). Controls for newcomer status and for co-authors' experience move it little (+0.48 and +0.50), though a confounder in co-author quality could also bias it: in a simulation reported in its declaration, such a confounder moved the estimate. Likewise, the same author is accepted about 1.4 pp less often [−2.7, −0.0] on a paper in a topic new to them (author fixed effects; EXPLORATORY).[2]

One topic insider on the team, against none, goes with +3.42 pp [+1.18, +5.66] in discovery and +3.88 [+0.28, +7.47] held-out, at fixed team size and number of experienced authors (HELD-OUT SUPPORTED). It does not survive the family-wise correction (Holm 0.052), and a status reading fits it as well: per head, an insider goes with only +0.58 pp more than an experienced author from another topic (p 0.16; EXPLORATORY).

The field's record is undetermined (uninformative). In the same space, the least-distant fifth of papers against the middle fifth is −3.27 [−4.93, −1.60] in discovery but −0.13 [−3.50, +3.25] held-out, with power 0.47 (uninformative), and the within-author slope for distance from the field was not supported either. Of the distance-based tests, only the own-record associations replicated.

Readings. These associations are referenced to a record, as continuity predicts, but expertise, evaluator conservatism and visible status predict them as well (Li, 2017; Merton, 1968). Double-blind review does not rule status out: where reviewers see the authors, famous authors fare better (Tomkins et al., 2017), and preprints can reveal who wrote a paper (Bharadhwaj et al., 2020). Pivots have been related to publication success (Hill et al., 2025); we observe the accept/reject decision itself, for every decided submission.

7 Discussion: Where Conformity Lives

What the evidence supports. The registered herding predictions failed at entry (follow-on work) or stayed inconclusive (flocking), and the registered crowding test at the gate was uninformative. Every association that replicated at the gate is referenced to the team's own record, except persisting topic

[2]Newcomer papers as a whole are accepted 10.2 pp less often [−11.8, −8.7] within topic and year (EXPLORATORY), but that gap mostly reflects who writes them (Appendix F).

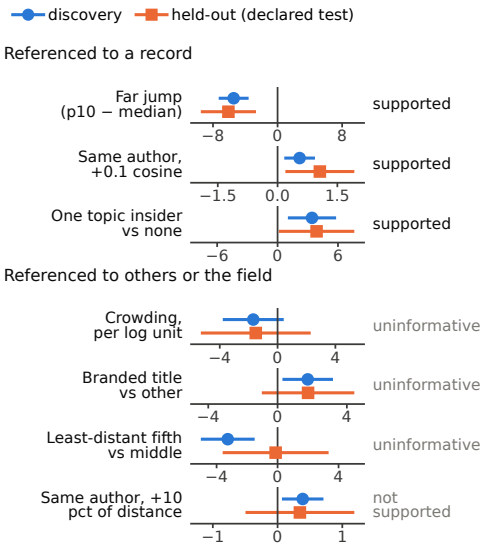


Figure 4: Gate tests by reference point: discovery estimates (blue circles) and held-out declared tests (orange squares), 95% intervals; the held-out verdict is at right (supported: HELD-OUT SUPPORTED; uninformative: not supported and below the test's MDE80; not supported: a declared test without a registered MDE80). Each row has its own scale: pp for the far jump, insider, branded title and least-distant fifth; pp per log unit for crowding; per 0.1 cosine and per 10 percentile points for the two same-author slopes. Compare replication status, not size. Crowding was registered; title style had been explored before registration; the other rows were found post hoc in discovery. Each held-out test was declared before any held-out estimate for that question was computed.

edges, which fit imitation of past acceptance and a rival (regression to the mean in noisy review) alike (Table 4). The conformity we do detect is a pull towards a record rather than towards identical work: incumbents' papers sit near their own earlier work, and acceptance goes with closeness to the team's record. But the record-side gate results do not single out continuity. Status and expertise predict them as well, and the test that could separate them, whether they concentrate where a preprint identifies the authors (Ibrahim et al., 2026), has not been run.

What remains open. Whether entrants crowd into hot topics is unsettled, and the absence of a detectable premium does not rule out bandwagons that bring no premium (Abrahamson and Rosenkopf, 1993); the spread of branded titles may be one, or it may carry a small premium; the held-out test cannot tell. Acceptance is also a short-run outcome: in the previous shift, researchers who

	Entry (what is written)	Gate (acceptance)
Field, both routes: convergence on the FM package (REPLICATED), also predicted by lock-in and efficient choice.		
Imitation (others, now)	Follow-on prediction reversed (REPLICATED); newcomers farther from earlier work; fewer twins for returning topic newcomers (EXPLORATORY); flocking inconclusive.	No detectable crowding premium (uninformative); branded titles: small premium, held-out uninformative (EXPLORATORY). Fit a rival too: edges partly kept (HELD-OUT SUPPORTED); no detectable change in the FM premium; timing weak (EXPLORATORY).
Continuity (a record)	Incumbents write more measurement papers (REPLICATED; much of it accounted for by team size); the package travels with an ICLR record, as diffusion also predicts (EXPLORATORY).	Lower far from the team's record; higher for the same author's closer work and with a topic insider (HELD-OUT SUPPORTED; status and expertise fit too).

Table 4: Where conformity shows, by route and site: a summary of Sections 4–6, with no new estimates. The field's record at the gate is uninformative.

stayed close to their own record had lower later impact (Bao and Li, 2025), so acceptance and impact need not go with the same work (cf. March, 1991; Rzhetsky et al., 2015).

Why the distinction matters. The two routes carry different risks. Imitation can discard private information and leave a fragile consensus (Bikhchandani et al., 1992; Banerjee, 1992); continuity trades long-run exploration for short-run competence (March, 1991). A field that looks herded may conform by continuity instead, and a diagnosis of fads would then misread it. For histories of NLP and ML (Hall et al., 2008; Anderson et al., 2012), an early-window refit and lagged predictors are a cheap way to see which trend claims survive hindsight.

8 Conclusion

We tested the herding account on a decade of ICLR submissions. At the field level, keywords rose and fell, emergence depends on hindsight, and the durable shift, the FM package, fits every route. At entry, researchers new to a topic bring methods, and, in the FM era, benchmark and evaluation work sits with incumbents and larger teams. At the gate, no premium for crowded topics is detectable, and acceptance goes with closeness to the team's own record. The conformity we can detect at ICLR points to a record, not to the crowd.

Limitations

All results are associations at one venue; none measures what choosing a topic, team, title or contribution type would change. The held-out set shares conference-years, reviewer pools and authors with the discovery papers, so it guards against analyst overfitting but not against year- or reviewer-specific patterns, and supported tests confirm direction, not size. The imitation–continuity frame was drawn up after the registered and declared tests had run; continuity's predictions were written afterwards. The record-side gate results were found post hoc, two of the three are secondary held-out tests, and the insider association does not survive a family-wise correction. Under one simulated null the main entry tests rejected 12.6% and 10.0% of the time at a nominal 5%. Contribution and FM labels are LLM labels, consistent across passes and hand-checked by the author on a small sample only; a method paper is a contribution type, not a measure of novelty, an adapter on an LLM counts as a method, and embedding distance is not originality. Entry status counts ICLR submissions only, and the 2021 switch from e-mail to profile identifiers hides earlier histories, which pulls entry contrasts towards zero. Abstracts and titles are partly camera-ready: links were removed and the text re-embedded for discovery papers only, and titles could not be corrected. Withdrawn papers are left out of acceptance tests. Status and continuity are not separated: work that continues a known line may itself reveal its authors, and we did not run the test of whether the record-side associations concentrate where a preprint identifies the authors. The hindsight comparison is a bracket, because the maps differ in fitting corpus as well as window. Keyword popularity and title style had been explored against acceptance before registration.

Ethics Statement

Author identifiers were used only as keys to compute entry status, closeness and the exclusion of a paper's own authors' earlier work; no author is named and all results are aggregates. The data are public OpenReview records, collected in our own dated acquisitions under the platform's terms. No review scores are used. Contribution labels were produced by an LLM (Claude Haiku) with a fixed codebook; agreement figures compare LLM labellings, not human judgement.

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